EPSR-test-ex113 Three-body potentials with ireset 1, more than 9 angles

Program(s): epsr | chained passes: 2 | verification: ALL IDENTICAL (all 2 passes)

What this case tests

Exercises the three-body input of the `epsr` engine [1,2,3] at its limits: a group with more angles than the 9 allowed, and a complete potential reset (`ireset 1`) in the same run.

Code exercised

Fortran (original, fortran_src/):

- `threebody_mod.f90`: `find_triplets`, `en_mole_triplets`, `pr_mole_triplets`, `save_threeb_energies`, `get_triplet_pot_cons`
- `read_and_save_files.f`: `read_pcof` (`tb` lines), `save_files_others` (re-reads and writes them)
- `ato_manipulation_rev4.f`: `fmolec1`, `press` (three-body energy/pressure); `molfit.f`: `en_change`

Rust (port, rust/epsr-core/src/):

- `threebody.rs`: `read_pcof_threebody`, `write_pcof_threebody`, `find_triplets`, `en_mole_triplets`, `pr_mole_triplets`, `save_threeb_energies`, `en_change_triplets`
- `energy.rs`: `fmolec1`, `en_change`, `press`; `pcof.rs`: `read_pcof`
- `pcof.rs`: `default_pair_potentials` (`ireset 1`)

Physical system

The EPSR-test system: a liquid mixture of 200 dimethyl sulphoxide (DMSO, (CH3)2SO) and 400 water molecules (DMSO mole fraction 1/3; 3200 atoms) at 300 K in a cubic box of side 33.12 Å (atomic number density 0.0881 atoms/Å³). DMSO uses EPSR's Lennard-Jones + Coulomb reference potential with methyl rotational groups; water uses SPC/E Lennard-Jones and charge parameters [4] with EPSR's flexible intramolecular geometry (O-H 0.976 Å). The empirical potential is refined against three neutron diffraction data sets of deuterated DMSO in D2O, in null-scattering H2O/D2O and in H2O (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), the isotope-substitution approach of Soper and Luzar [5]. A `tb A1 A2 A3 amp awidth rwidth rmax12 rmax32 s12 s23 s13 ang1 amp1 ...` line in the `.pcof` adds an angular energy `amp*ampN*0.5*((cos theta - cosN)/awidth)²` to every triplet A1-A2-A3 (A2 the centre) within the distance limits, for the requested angle N closest to theta; the flags select same or different molecules. Groups: water O1-H1···O1with 10 angle/amplitude pairs (180° to 135°) plus an odd trailing word, and intramolecular DMSO O-S-C (106°).

Inputs

EPSR-test with the two `tb` lines appended to the `.pcof`; `ireset 1` in the `.EPSR.inp` (reset to 0 on saving, so pass 2 runs with `ireset 0`). `commands.txt`: `epsr 200DMSO400Water.EPSR.inp`, `repeat 2`.

How the accuracy was verified

`commands.txt` (1 program call, repeated 2 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 2 passes.
`(nwords-12)/2` = `10` angles are capped at 9. Pass 2 reads back the `.pcof` written by pass 1, with the altered `tb` values described below, so both the reset path and the re-read of the program's own output are compared.

Notes

Fortran quirk reproduced: with `ireset 1` the default pair potentials reset `nsymbol`, so the `tb` symbols are no longer counted; when the `ndata` line is processed the `tb` values are not moved and at save time the groups are not re-read: the in-memory values are written (width 1.0 rad -> `57.30`, cosines -> angles such as `-57.30`), and pass 2 uses them.

References

[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[4] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[5] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.